

# ML Regression Project: Life Expectancy Prediction

## Dataset: Life Expectancy Data

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### 1. Introduction

Healthcare and population health analytics play a pivotal role in understanding the overall well-being and development of nations. Among various public health indicators, **life expectancy** is one of the most significant, as it reflects a country's socioeconomic conditions, quality of healthcare systems, educational attainment, and overall disease burden.

### 1. Objective

The objective of this project is to build a Machine Learning regression model to predict Life Expectancy using health-related indicators such as Adult Mortality, Infant Deaths, BMI, and GDP.

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### 2. Dataset Description

The dataset used is 'Life Expectancy Data.csv'. It contains country-level health, economic, and demographic indicators collected over multiple years.

### 3. Features and Target

**Type**

**Columns**

Features (X)

Adult Mortality, infant deaths, BMI, GDP

Target (y)

Life Expectancy

### 4. Machine Learning Model

A Multiple Linear Regression model was used. The dataset was split into training and testing sets (80% training and 20% testing). The model learns the relationship between the selected features and life expectancy.

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## 5. Model Evaluation

The model performance was evaluated using  $R^2$  Score and Mean Absolute Error (MAE). The achieved  $R^2$  score of approximately 0.64 indicates that the model explains around 64% of the variance in life expectancy. The MAE of about 4 years shows reasonable prediction accuracy for real-world health data.

## 6. Coding

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error

# =====
# Load Dataset
# =====
df = pd.read_csv("Life Expectancy Data.csv")

print("Dataset loaded successfully ✔")

# =====
# Clean column names
# =====
df.columns = df.columns.str.strip()

print("\nCleaned Columns:\n", df.columns)

# =====
# Required Columns
# =====
required_columns = [
    'Life expectancy',
    'Adult Mortality',
    'infant deaths',
    'BMI',
    'GDP'
]

df = df[required_columns]
df = df.dropna()
```

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```
# =====  
# Features & Target  
# =====  
X = df[['Adult Mortality', 'infant deaths', 'BMI', 'GDP']]  
y = df['Life expectancy']  
  
# =====  
# Train Test Split  
# =====  
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)  
  
# =====  
# Train Model  
# =====  
model = LinearRegression()  
model.fit(X_train, y_train)  
  
# =====  
# Prediction  
# =====  
y_pred = model.predict(X_test)  
  
# =====  
# Evaluation  
# =====  
print("\nModel Performance:")  
print("R2 Score:", r2_score(y_test, y_pred))  
print("MAE:", mean_absolute_error(y_test, y_pred))
```

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The screenshot shows a Visual Studio Code editor with a file named `ML-regression.py` open. The script contains the following code:

```
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53 # =====
54 # Prediction
55 # =====
56 y_pred = model.predict(X_test)
57
58 # =====
59 # Evaluation
60 # =====
61 print("\nModel Performance:")
62 print("R2 Score:", r2_score(y_test, y_pred))
63 print("MAE:", mean_absolute_error(y_test, y_pred))
64
```

The terminal output shows the execution of the script:

```
PS C:\Users\Vanisree\.vscode\New folder> & C:/Users/Vanisree/AppData/Local/Microsoft/WindowsApps/python3
.exe "c:/Users/Vanisree/.vscode/New folder/ML-regression.py"
Dataset loaded successfully ✓

Cleaned Columns:
Index(['Country', 'Year', 'Status', 'Life expectancy', 'Adult Mortality',
      'infant deaths', 'Alcohol', 'percentage expenditure', 'Hepatitis B',
      'Measles', 'BMI', 'under-five deaths', 'Polio', 'Total expenditure',
      'Diphtheria', 'HIV/AIDS', 'GDP', 'Population', 'thinness 1-19 years',
      'thinness 5-9 years', 'Income composition of resources', 'Schooling'],
      dtype='object')

Model Performance:
R2 Score: 0.6426674455657923
MAE: 3.9778441699189915
PS C:\Users\Vanisree\.vscode\New folder>
```

## 7.Conclusion

This project demonstrates the application of machine learning regression techniques to real-world health data. The model provides meaningful insights into how mortality, economic conditions, and health indicators affect life expectancy.